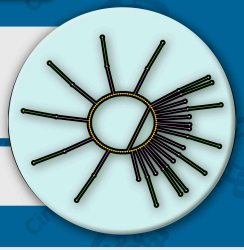
General Attraction Diagrams for Cellular Automata

The **general attraction diagram** represents all the basins of attraction for all possible cellular automata with length n and a given progression rule.

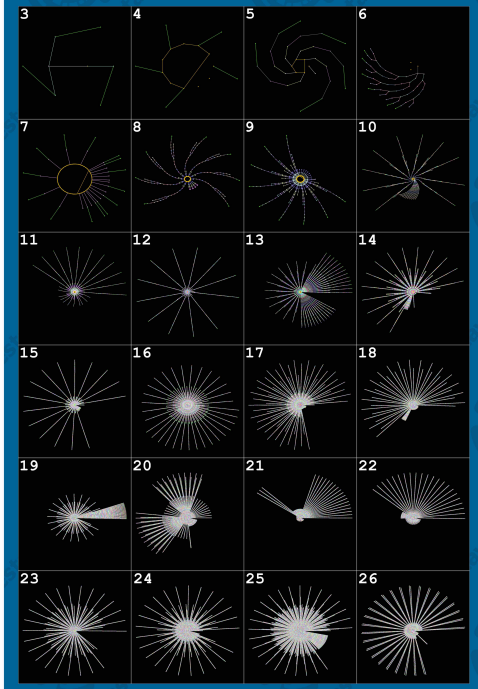
Green nodes (Gardens of Eden) cannot be obtained through progression.

Yellow nodes are in the attractor.

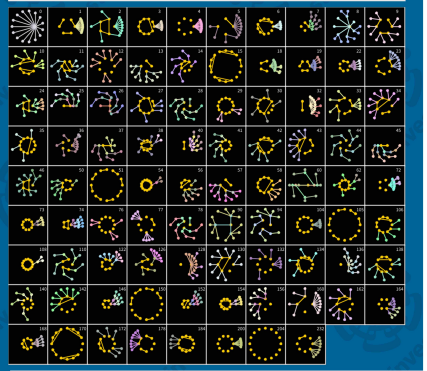


For this diagram n = 7 Rule 30
Chaotic dynamic

Observe diagrams for Rule 30 in different lengths



Observe diagrams for different rules with n=4



The diagrams describe the **global dynamics** of the system (chaotic, complex, or ordered). This is a heuristic for the entropy of the system. Each rule represents **different data distribution** for data science.

Different configurations of data are represented as nodes. Note how some diagrams group data into long or short branches, with **long or short cycles**, and with clustered or sparse data.

Rule 30, known for its applications in random number generation, **is chaotic**, and its diagram features sparse, long branches and long attraction cycles. Meanwhile, **Rule 22**, known for generating **emergent behavior**, has abundant, short branches and short attraction cycles.

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FEW-AI 2025

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